

Assignment-3:

Prove that the set of rational numbers $\mathbb{Q}$, equipped with two binary operations of addition and multiplication forms a field.

Ans: A field is a set F with two binary operation $(+)$ and $(\cdot)$ satisfying following properties:

1. $(F, +)$ is an abelian group

2. $(F \setminus \{0\}, \cdot)$ is an abelian group

3. Distributive law: for all $x, y, z \in F$,
 $x \cdot (y + z) = x \cdot y + x \cdot z$

1. Additive group properties

i) closure under $+$:

Let $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ with $b, d \neq 0$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

Since $ad + bc \in \mathbb{Z}$ and $bd \in \mathbb{Z} \setminus \{0\}$ the sum is a rational number. So, $\mathbb{Q}$ is closed under addition.

ii) Associativity and commutativity of $+$:

These follow from the associativity and commutativity of integer addition and of the formula for adding fraction.

iii) Additive identity: $0 = \frac{0}{1} \in \mathbb{Q}$ and for any $\frac{a}{b}$: $\frac{a}{b} + 0 = \frac{a}{b}$

iv) Additive inverses: For $\frac{a}{b} \in \mathbb{Q}$ is additive inverse is $-\frac{a}{b} = \frac{-a}{b} \in \mathbb{Q}$ because

$$\frac{a}{b} + \frac{-a}{b} = \frac{0}{b} = 0$$

Thus, $(\mathbb{Q}, +)$ is an abelian group.

2. Multiplicative group properties:

i) closure under:

$$\text{For } \frac{a}{b}, \frac{c}{d} \in \mathbb{Q}, \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

and $ac \in \mathbb{Z}$, $bd \in \mathbb{Z} \setminus \{0\}$, so the product is in $\mathbb{Q}$

ii) These follow from associativity and commutativity of integer multiplication.

iii) multiplicative identity: $1 = \frac{1}{1} \in \mathbb{Q}$ and

for any $\frac{a}{b}$, $\frac{a}{b} \cdot 1 = \frac{a}{b}$

iv) multiplicative inverses: If $\frac{a}{b} \in \mathbb{Q}$ and $\frac{a}{b} \neq 0$ then $a \neq 0$. Its inverse is $\frac{b}{a} \in \mathbb{Q}$

because, $\frac{a}{b} \cdot \frac{b}{a} = \frac{ab}{ba} = 1$

v) Non zero identity:

$0 = \frac{0}{1}$ and $1 = \frac{1}{1}$; clearly $0 \neq 1$

Hence $\mathbb{Q} \setminus \{0\}$ is an abelian group.

3. Distributive law:

For rational $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$

$$\begin{aligned} \frac{a}{b} \left(\frac{c}{d} + \frac{e}{f} \right) &= \frac{a}{b} \cdot \frac{cf + de}{df} = \frac{a(cf + de)}{bdf} \\ &= \frac{acf}{bdf} + \frac{ade}{bdf} = \frac{a}{b} \cdot \frac{c}{d} + \frac{a}{b} \cdot \frac{e}{f} \end{aligned}$$

All field axioms are satisfied. So, $\mathbb{Q}$ with usual addition and multiplication is a field.